MILITARY SPECIFICATION

COATING PLASTIC, DENTURE MODEL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers coating plastic suitable for use in preparing denture models.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

STANDARDS

Federal

FED-STD-123	Marking For Domestic Shipment (Civilian Agencies).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:51Z
Check the source to verify that this is the current version before use.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout and shall be suitable in all respects for the intended purpose.

3.1.1 Coating plastic. The coating plastic shall be composed essentially of acrylic resins, together with suitable modifying resins, plasticizers, and solvents to meet all the requirements of this specification. The propellant shall be a hydrocarbon, a halogenated hydrocarbon, or a blend of either or both hydrocarbons and halogenated hydrocarbons.

3.1.2 Dispenser. The container shall be fabricated from suitable steel. The cover cap shall be fabricated from metal. Components of the valve assembly shall be fabricated from suitable materials which will not react with the coating plastic or the propellant.

3.2 Design and construction. The dispenser shall consist of a container, a dispensing valve, and a cover cap.

3.2.1 Container. The container shall be a pressurized aerosol can with a capacity sufficient to hold a minimum of 12 ounces of the coating plastic. The container shall have a round, drawn, seamless body and a domed neck with a one-inch-diameter opening. The bottom shall be concave and shall be compound lined and double seamed to the body. The opening in the dome shall be properly fitted with the valve assembly specified in 3.2.2.

3.2.2 Valve assembly. The valve assembly shall consist essentially of an actuator, a stem, a gasket, a spring, a mounting cup with flowed-in gasket, the housing, and dip tube. The mounting cup shall have a one-inch diameter, and shall properly fit the dome at the top of the container. The actuator shall be removable from the stem without the release of pressure from the aerosol. A metering orifice in the bottom of the actuator shall be accessible for cleaning with a straight pin, upon removal of the actuator from the stem. An orifice in the side of the actuator shall be of such dimensions as will produce the spraying properties required in 3.6. The lower end of the dip tube shall have a maximum clearance of 1/4 inch from the bottom rim of the can, after storage for at least one week in the product (to allow for swelling of the tube).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:51Z
Check the source to verify that this is the current version before use.

3.2.3 Cover cap. A removable and replaceable friction-fitted cover cap shall completely cover the valve assembly and shall protect the actuator from accidental functioning and possible damage.

3.2.4 Coating plastic. The coating plastic (prior to dispensing) shall be in liquid form, and shall be suitable for coating and filling the pores of investment models to provide a smooth, glazed surface prior to the application of plastic patterns in the construction of partial dentures. The liquid shall be under sufficient pressure to permit spraying of the entire contents. The pressure shall not exceed 75 pounds/square inch (gage) at 70° F, when tested as specified in 4.4.1. The application of thumb pressure to the top of the spray head shall enable the liquid to be dispensed directly onto the models.

3.3 Net contents. The net contents of each unit can shall be 340 ± 5 grams by weight, when determined as specified in 4.4.2.

3.4 Color. The coating plastic shall be clear and colorless.

3.5 Total solids. The percentage of total solids in the coating plastic shall be 3.0 to 5.0 percent, when determined as specified in 4.4.3.

3.6 Spraying properties. The aerosol shall deliver a spray of normal particle size; overspray shall not be excessive; the spray pattern shall be such as to deposit a uniform, bubble-free, adherent coating on an investment model surface; and the film shall dry free from tackiness, orange peel, sags, specks, or any other surface irregularity, when tested as specified in 4.4.4. The valve shall operate freely without excessive finger pressure and shall close immediately upon release of finger pressure; the spray shall be continuous without sputtering or interruption; and the valve shall not clog nor collect heavy deposits of coating around the orifice, when tested as specified in 4.4.4.

3.7 Delivery schedule. No more than two months shall have elapsed from date of manufacture of the coating plastic to date of delivery to the Government.

3.8 Finish. The can and cover shall be enamel coated or otherwise treated to prevent corrosion of the metal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:51Z
Check the source to verify that this is the current version before use.

3.9 Instructions. Adequate and accurate instructions for use shall appear on the label securely affixed to the container body of each unit can. Instructions shall essentially be as follows:

DIRECTIONS

TO OPERATE: Lift protective cap. To spray - push down on button. Hold dispenser 6" to 8" from model. Aim at design and move across area to be sized. Cover hand holding model with a towel to avoid getting spray on hand.
FOR GOLD CASTING: Remove investment model from duplicating material; design; then spray working surface sparingly until a gloss appears. (Model need not be Rosin Dipped.) Place in (suggested manufacturer) Drying Oven at 200° F. until thoroughly dry. Warm model and apply (suggested manufacturer) "Pre-Formed" Plastic Patterns. Soak model well before investing.
FOR CHROME CASTING: Dry and dip refractory model in (suggested manufacturer) Rosin Dip; design; then spray model lightly over designed area. Apply (suggested manufacturer) "Pre-Formed" Plastic Patterns and invest as recommended by investment manufacturer.

3.10 Workmanship. Workmanship shall be first class throughout. The coating plastic and each unit can shall be free from defects which affect their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:51Z
Check the source to verify that this is the currenversion before use.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables I and III (see 4.2.1 and 4.3) and table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be a unit can of coating plastic.

Table I. Examination criteria

Type defect	Inspection level	AQL (Percent defective)
Major	II	1.5
Minor	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one unit can of the coating plastic.

Table II. Test criteria

Characteristic	Requirement	Test method	Inspection level
Pressure test	3.2.4	4.4.1	S-2
Net contents	3.3	4.4.2	S-1
Total solids	3.5	4.4.3	/1
Spraying properties	3.6	4.4.4	S-1

/1 A sufficient amount from each batch or lot shall be selected for the determination of total solids.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:51Z
Check the source to verify that this is the current version before use.

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used in the determination of two or more test characteristics.

4.3 Examination. Examination of the coating plastic and the unit can shall be made in accordance with the following classification of defects, but shall not be restricted thereto.

Table III. Classification of defects

Category	Defects
Major:	
101	Coating plastic not clear and colorless.
102	Any component of dispenser missing.
103	Design and construction of dispenser not as specified.
Minor:	
201	Cover cap does not properly fit over the valve assembly.
202	Instructions for use not as specified.
203	Packaging or packing not as specified.
204	Marking on packaging and pack not as specified.

4.4 Tests.

4.4.1 Pressure test. The filled can shall be tested with a calibrated pressure gauge to determine the internal pressure of the can.

4.4.2 Net contents. The cover cap shall be removed from the filled dispensing can. The can with contents shall be weighed to the nearest gram on a suitable balance. The plastic coating shall be completely sprayed out under proper ventilation until no more coating remains. The can shall then be weighed again to the nearest gram. The difference in weight in grams between the can before and after dispensing shall be the net contents of the unit can.

4.4.3 Total solids. The filled dispensing can shall be refrigerated for one hour. The can shall then be agitated well, and the top of the can shall be pierced carefully with a needle point to allow the pressurized propellant to escape. After a few minutes, the pierced hole shall be enlarged, and as soon as there is no further evidence of propellant escaping, the container shall be cut open and the contents transferred into a tared glass weighing bottle, or vessel , and weighed to the nearest gram. The solution in the weighing bottle or vessel shall be allowed to evaporate at room temperature of 70° F. ± 5° F. for 24 hours, followed by heating in a 100° F. ± 5° F. drying oven for 1 hour. The weighing bottle or vessel shall then be allowed to cool to room temperature and shall be weighed. The percentage of total solids shall be calculated to the nearest 0.1%.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:51Z
Check the source to verify that this is the currentversion before use.

4.6.4 Spraying properties. A dry dental investment shall be sprayed at 70° F. ± 5° F. room temperature, in accordance with operating instructions on unit can. The model shall be immediately transferred to a 100° F. ± 5° F. oven for 12 seconds. Next, the model shall be removed to 70° F. ± 5° F. room temperature. A dental resin pattern shall be applied to the model. The pattern shall then be removed, and the model examined for evidence of coating coming off with the pattern. The spray and the sprayed model shall comply with the properties specified in 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One can, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
Not Required	12 units	24 units

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Intermediate package. Intermediate package shall be a box of appropriate size and design, constructed in accordance with PPP-B-636, type CF, class weather-resistant. Carton design shall include full or shoulder height, half-slotted style partitions. Partitions shall be fabricated of double-faced or double-wall fiberboard having a minimum bursting strength test of 200 pounds. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with either adhesive (conforming to MMM-A-250) or metal stitches is permitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:51Z
Check the source to verify that this is the current version before use.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table III, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table II, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:51Z
Check the source to verify that this is the current version before use.

5.3.3 Special marking. Legible commercial marking shall be acceptable. It shall be lithographed on the unit container or printed on a label securely affixed to the container and shall include the following:

(a) Federal Stock Number.
(b) Quantity and unit.
(c) Item identification.
(d) Manufacturer's name and address.
(e) Precautions for use and storage.
(f) Directions for use, (see 3.9).
(g) Contract or purchase order number.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type which is normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:51Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:51Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-C-36584 FSN

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 20261O-21T08:51Z
Check the source to verify that this is the current version before use.

CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

Fold

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

DEFENSE SUPPLY AGENCY
OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

Fold

DPSC FORM 2418
MAR 68

EDITION AUG 66, WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:51Z
Check the source to verify that this is the current version before use.